pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

Case Study: Joining the Tables

WEEK 2 · DAY 5

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Three clean tables, and one book to build

What yesterday found

Table	Rows	Columns	One row is
`positions`	200	6	one bond held
`issuers`	150	5	one issuer
`ratings`	18	4	one rating

489,000,000
of face held in total

0
missing values or duplicates

2
keys tested: ticker and rating

The report to the desk: the three tables are clean and they fit together.

Your job today

Build — one table, the book: one row per bond, with every attribute the desk needs

Prove — that nothing was lost and nothing was duplicated on the way

Answer — the desk's nine questions from that one table

Two joins, and a check around each

`positions` **merge on ticker** `issuers` → **step_one**

Gives each bond its issuer's name, sector, country, and rating.

`step_one` **merge on rating** `ratings` → **book**

Gives each bond its score, its bucket, and its grade.

The check

Written down before any join:

200 rows

489,000,000 of total face

After each join, both must be unchanged.

```
step_one = pd.merge(positions, issuers, on='ticker', how='left', validate='many_to_one')
book = pd.merge(step_one, ratings, on='rating', how='left', validate='many_to_one')
# how='left' keeps every row of the left table. validate= stops the merge if a key is listed twice in the lookup.

assert book.shape[0] == rows_before, 'the second merge changed the row count'
assert book['par_held'].sum() == face_before, 'the second merge changed the total face'
# assert stops the notebook with its message if a condition is False
```

The same join in Excel

a lookup, filled down the sheet

`=VLOOKUP(B2, issuers!A:E, 3, FALSE)`	the sector: column 3 of A to E
`=XLOOKUP(B2, issuers!A:A, issuers!C:C)`	the sector, in current Excel

The book: 200 rows and 13 columns

200
rows, after both joins

13
columns: 6, plus 4, plus 3

489,000,000
of face, unchanged

0
gaps in the new columns

```
book.iloc[0]
cusip                99080CAE8
ticker                    BEZE
coupon                     7.0
maturity            2032-06-18
price                  106.727
par_held               5000000
issuer          Bezane Finance
sector               Financials
country                      US
rating                     BBB-
score                        10
bucket                      BBB
grade         Investment Grade
```

The check after the second join. Both assert lines passed in silence.

```
Rows before: 200  after: 200
Face before: 489,000,000  after: 489,000,000
Rows unchanged: True
Face unchanged: True
```

Before the second join

`step_one.groupby('grade')`

KeyError: 'grade'. The column is not in the table yet.

→

After it

`book.groupby('grade')`

It runs. The grade came from the ratings table.

An outer join keeps every key from both tables

```
everything = pd.merge(positions, issuers, on='ticker',
                      how='outer', indicator=True)
everything['_merge'].value_counts()

both          200
right_only     35
left_only       0
```

200 both: a position that found its issuer

35

235 rows. 35 right_only: an issuer with no position

These 35 rows are not errors in the data. They are issuers the desk does not hold, and they do not belong in a table with one row per bond held.

What a right_only row looks like

ticker	issuer	cusip	par_held
BEZY	Bezolvey Chemicals	NaN	NaN
BRUN	Bruxiskan Pipeline	NaN	NaN

A report built on this table by mistake

489,000,000

of face. The face check passes: sum() skips the missing values.

235

rows, against 200. The row check fails. As an assert, it stops the run:

```
AssertionError: expected 200 rows and found 235
```

One check passes and one fails. Always run both.

A repeated key adds rows, with no error

Four positions, and a lookup that lists BEZL twice

```
doubled = pd.merge(small, lookup_twice, on='ticker', how='left')

       cusip ticker  par_held          sector
0  99080CAE8   BEZE   5000000      Financials
1  99049XAE2   BEZH    250000     Industrials
2  99039NAC0   BEZL   1500000  Transportation
3  99039NAC0   BEZL   1500000  Transportation
4  99039NAG1   BEZL    500000  Transportation
5  99039NAG1   BEZL    500000  Transportation

Face before: 7250000  after: 9250000
```

6 rows

from 4: each BEZL bond matched twice

9,250,000

of face, from 7,250,000

The same lookup sheet in Excel

VLOOKUP and XLOOKUP return the first match and stop. No row is added, and the repeat is not reported. pandas returns every match.

Join written as	Rows	Face	Caught by
left, each key once in the lookup	200	489,000,000	nothing to catch
outer	235	489,000,000	the row check
left, a key repeated in the lookup	more than the left table	too high	**both checks, and validate= before them**
inner, a key missing from the lookup	fewer than the left table	too low	**both checks**

Preparing the book: five new columns

Column	pandas	Excel, row 2
Value	`book['par_held'] * book['price'] / 100`	`=F2 * E2 / 100`
Coupon income	`book['par_held'] * book['coupon'] / 100`	`=F2 * C2 / 100`
Years to maturity	`(book['maturity'] - as_of).dt.days / 365.25`	`=(D2 - DATE(2026, 9, 30)) / 365.25`
Weight	`book['value'] / total_value * 100`	`=N2 / SUM($N$2:$N$201) * 100`

Value at the clean price

`face x clean price / 100`

The first bond: 5,000,000 of face at 106.727 is 5,336,350. This is before accrued interest, so it is not a full market value.

A date, then a bucket

`pd.to_datetime()` `pd.cut()`

The maturity arrived as text. As a date it gives the years left, and pd.cut() sorts those into 0 to 3, 3 to 7, 7 to 15, and 15 and over.

Helper columns in Excel

`$N$2:$N$201`

Each new column is a formula filled down the sheet. The dollar signs hold the total in place. In pandas the total is one stored number.

The desk's questions: sector, grade, rating

Bonds and value by sector, largest first

sector	bonds	value	pct_of_value
Materials	26	70,527,440	14.60
Financials	26	67,067,357.5	13.88
Consumer	26	55,773,490	11.55
Telecom	21	54,953,580	11.38
Industrials	20	51,658,930	10.69
Transportation	20	48,159,510	9.97
Health Care	20	38,268,482.5	7.92
Energy	17	38,258,305	7.92
Technology	12	29,826,430	6.17
Utilities	12	28,530,705	5.91

In Excel: a pivot table with sector in Rows, and Count of cusip and Sum of value in Values.

The two grades: share of bonds and share of value

grade	bonds	pct_of_bonds	pct_of_value
High Yield	72	36.0	34.45
Investment Grade	128	64.0	65.55

In Excel, with the grade in column M and the value in column N

```
=COUNTIF(M:M, "High Yield")        72 bonds
=SUMIF(M:M, "High Yield", N:N)     166,398,102.5
```

Percent of value by rating bucket, sorted on the score

AAA	AA	A	BBB	BB	B	CCC
1.68	2.68	30.38	30.82	23.09	8.83	2.53

The desk's questions: averages, issuers, maturity

Average coupon, in percent, and average rating score

Average	coupon	score
Simple	5.9375	9.36
Value-weighted	5.9403	9.25

In Excel, with the coupon in column C and the value in column N

```
=AVERAGE(C2:C201)
```

```
=SUMPRODUCT(C2:C201, N2:N201) / SUM(N2:N201)
```

483,024,230

of value at the clean price, before accrued interest. The answers describe the data and are not a view.

Value by sector and grade: three rows of the pivot table

sector	High Yield	Investment Grade
Materials	38,003,020	32,524,420
Telecom	27,344,940	27,608,640
All	166,398,102.5	316,626,127.5

In Excel: the same pivot table, with grade dragged into Columns.

20.67%

ten largest of 115 issuers

46.61%

in the 7 to 15 year bucket

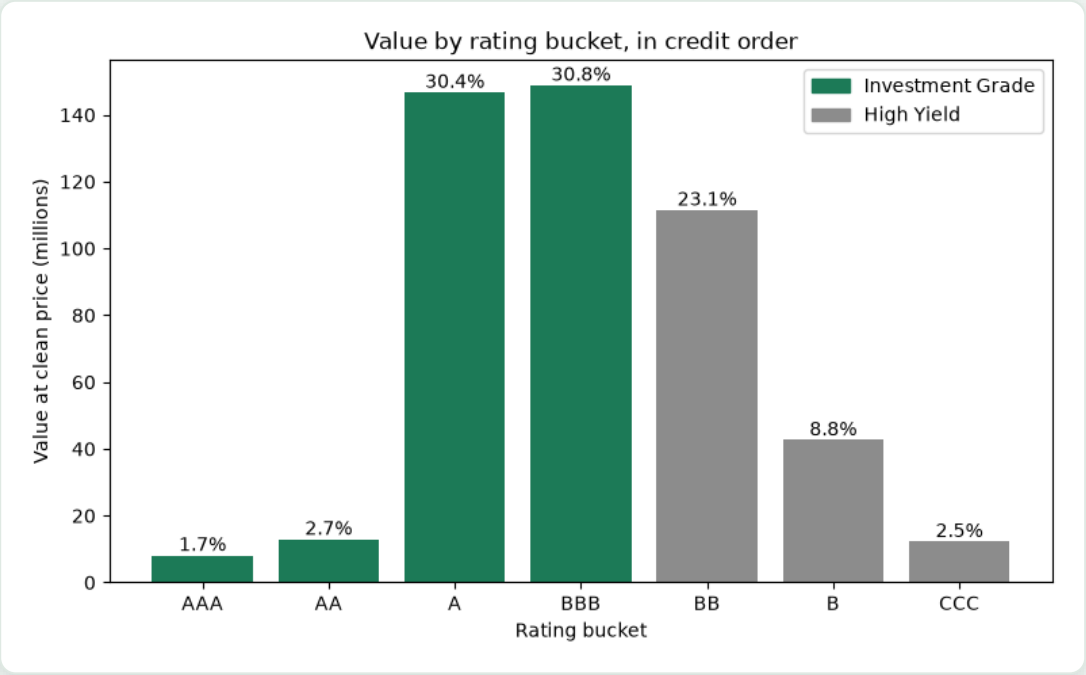
113

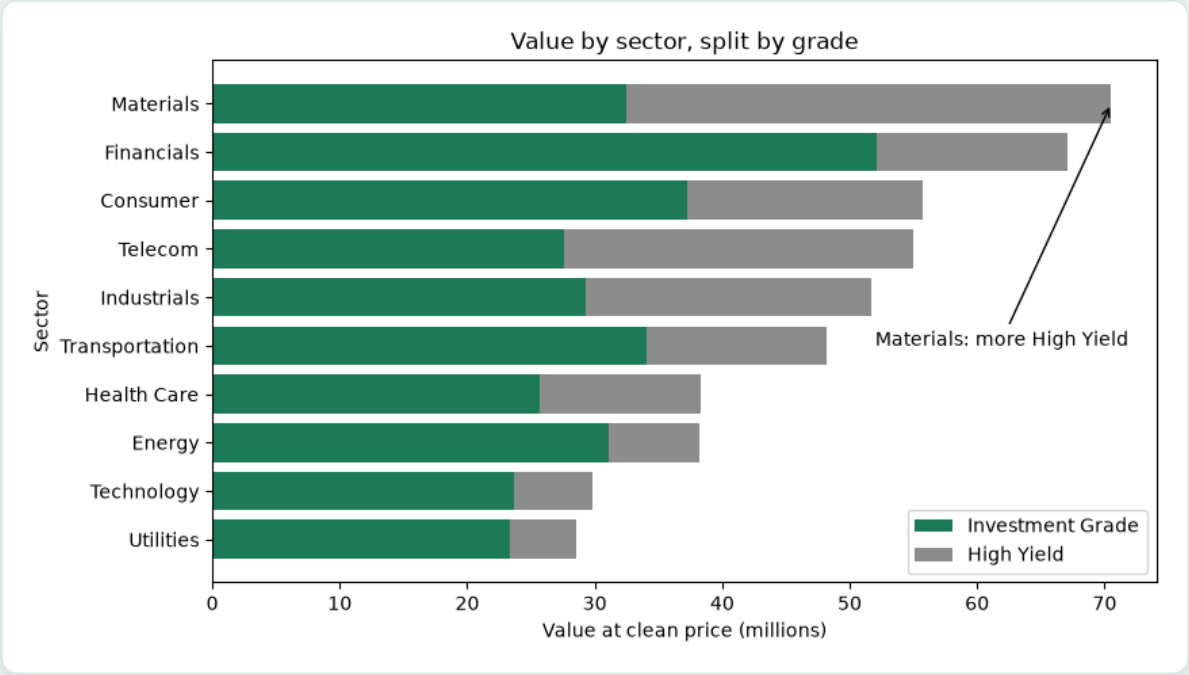
bonds priced below par

28

bonds mature within 3 years

Two pictures: rating buckets and sectors by grade





BBB and A are about 30 percent of the book each

Materials is the one sector with more High Yield than Investment Grade

Week 3 teaches charts. For now, read each picture and check it against its table.

Save the book, then read it back

To a CSV file, and back

```
book.to_csv('desk_book_joined.csv', index=False)
read_back = pd.read_csv('desk_book_joined.csv',
                        parse_dates=['maturity'])

Shape saved: (200, 18)  read back: (200, 18)
Value saved: 483,024,230  read back: 483,024,230
Face saved: 489,000,000  read back: 489,000,000
```

To an Excel workbook with two sheets, and back

```
with pd.ExcelWriter('desk_book_joined.xlsx') as writer:
    book.to_excel(writer, sheet_name='book', index=False)
    by_sector.to_excel(writer, sheet_name='by_sector')
from_excel = pd.read_excel('desk_book_joined.xlsx',
    sheet_name='book', dtype={'cusip': str})

Shape saved: (200, 18)  read back: (200, 18)
Value saved: 483,024,230  read back: 483,024,230
```

Column	In the book	From the CSV	From the workbook
maturity	datetime64[us]	datetime64[us]	datetime64[us]
maturity_bucket	category	str	str

Both files return the bucket as text, so run pd.cut() again. Older versions of pandas show object where this says str.

What each file keeps

A CSV holds characters only, so parse_dates asks for the date. A workbook stores a date as a date.

The check, once more

Shape, value, and face match after each read, and every assert line passes.

The same in Excel

File, Save As: CSV for one table, or Excel Workbook with a book tab and a by_sector tab.

The AI check

Every notebook ends with code written in the style of an AI assistant. It has one bug. Find it.

1

Read

Read the code before you run it.

2

Predict

Decide what it should produce.

3

Run

Run it and look at the output.

4

Check

Compare with a number you worked out yourself.

Code that runs is not the same as code that is right.

What you can do now

- Join tables on a key with pd.merge(), and how='left' to keep every row
- Check the rows and a total around every join, with validate= and assert
- Read _merge from an outer join to see what did not match
- Prepare new columns, with the unit in the name
- Answer by group with groupby(), agg(), and pivot_table(), and style the result

Week 3 Charts and exploratory data analysis: the same book, and the same questions, as pictures